

BAIE-COMEAU, QC, Canada

WMO: 716910

Lat: 49.1351N

Lon: 68.2024W

Elev: 22

StdP: 101.07

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.4	(c) -23.8	(d) -33.6	(e) 0.2	(f) -25.9	(g) -30.8	(h) 0.2	(i) -23.3	(j) 13.2	(k) -3.3	(l) 11.8	(m) -4.9	(n) 3.9	(o) 270	(p) 0.730

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	24.1	17.5	22.6	16.8	21.2	16.1	18.9	22.4	17.8	21.2	16.9	20.0	5.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.5	12.5	20.5	16.5	11.7	19.3	15.6	11.1	18.3	53.8	22.4	50.4	21.2	47.6	20.1	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.4	10.1	8.9	-30.8	28.1	2.6	1.6	-32.6	29.3	-34.2	30.2	-35.6	31.1	-37.5	32.3
			-31.0	21.2	2.6	1.2	-32.8	22.0	-34.3	22.7	-35.8	23.4	-37.6	24.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-12.6	-11.5	-6.0	0.9	7.4	12.9	16.2	15.4	10.9	5.2	-0.8	-7.8
	DBStd	10.83	6.69	6.38	5.66	3.52	3.07	3.01	2.22	2.40	3.10	3.21	4.45	5.80
	HDD10.0	3220	700	601	495	272	93	7	0	1	25	152	324	551
	HDD18.3	5749	958	834	753	522	338	163	74	93	222	407	574	809
	CDD10.0	524	0	0	0	0	13	95	191	169	53	3	0	0
	CDD18.3	12	0	0	0	0	0	2	7	4	0	0	0	0
	CDH23.3	83	0	0	0	0	2	18	38	23	3	0	0	0
	CDH26.7	7	0	0	0	0	0	2	3	1	0	0	0	0
Wind	WSAvg	4.1	4.4	4.5	4.5	4.3	4.0	3.9	3.7	3.6	3.8	4.2	4.3	4.4
Precipitation	PrecAvg	1045	88	71	76	81	89	83	78	91	102	89	98	101
	PrecMax	1301	198	140	160	208	187	185	146	196	166	134	177	216
	PrecMin	815	23	10	20	9	11	31	32	37	46	26	50	45
	PrecStd	151	40	33	39	48	42	35	33	39	36	30	36	41
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	3.4	6.0	11.9	21.5	25.6	26.4	25.9	22.7	16.3	10.3	4.7
		MCWB	1.7	1.9	2.3	6.5	13.9	16.8	19.0	18.4	17.1	11.8	8.7	3.8
	2%	DB	1.2	1.8	3.7	8.9	18.3	22.8	24.2	23.5	19.7	13.6	7.8	2.9
		MCWB	0.5	0.6	1.2	5.0	11.6	15.6	17.9	17.5	15.2	10.6	6.1	1.8
	5%	DB	-0.5	-0.1	2.4	7.0	15.2	20.8	22.8	21.9	17.8	11.6	6.2	1.7
		MCWB	-1.3	-1.1	0.6	3.6	10.0	14.7	17.2	16.9	13.9	9.1	4.9	0.8
	10%	DB	-2.5	-1.7	1.4	5.6	12.9	19.0	21.3	20.4	16.1	10.1	5.0	0.5
		MCWB	-3.3	-2.7	-0.1	2.8	8.8	13.8	16.4	16.0	13.0	7.9	3.9	-0.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.3	2.4	3.3	7.5	15.0	18.8	20.9	19.7	18.2	13.1	9.2	4.0
		MCDB	2.8	3.1	4.8	10.8	19.3	22.7	24.3	23.5	21.0	14.8	9.9	4.2
	2%	WB	0.6	0.7	1.8	5.6	12.2	17.1	19.2	18.5	16.0	11.2	6.5	2.1
		MCDB	1.2	1.4	3.0	8.0	16.6	21.0	22.7	22.0	18.8	13.1	7.2	2.6
	5%	WB	-1.3	-0.9	0.9	4.2	10.8	15.8	18.1	17.6	14.6	9.7	5.1	0.9
		MCDB	-0.6	0.0	2.1	6.4	14.1	19.5	21.6	20.7	16.9	11.2	5.9	1.5
	10%	WB	-3.3	-2.6	0.0	3.2	9.4	14.5	17.1	16.7	13.5	8.4	3.9	-0.1
		MCDB	-2.5	-1.8	1.2	4.9	12.2	17.7	20.3	19.5	15.6	9.6	4.7	0.5
Mean Daily Temperature Range	5% DB	MDBR	9.1	9.5	8.5	6.9	8.6	9.4	9.3	9.3	8.8	7.2	6.3	7.7
		MCDBR	9.0	7.0	7.4	9.1	13.7	13.5	12.1	11.9	11.1	9.5	6.6	6.2
		MCWBR	8.9	6.6	6.1	6.4	8.4	7.5	6.8	6.8	7.3	6.8	5.8	6.2
	5% WB	MCDBR	8.8	6.5	6.5	8.2	11.5	11.4	10.8	10.1	9.8	8.5	6.0	6.2
		MCWBR	8.8	6.3	5.8	6.1	7.7	7.1	6.8	6.3	7.1	6.6	5.7	6.2
Clear-Sky Solar Irradiance	taub	0.267	0.274	0.305	0.360	0.373	0.398	0.425	0.403	0.378	0.333	0.313	0.271	
	taud	2.342	2.318	2.292	2.268	2.340	2.347	2.304	2.367	2.411	2.512	2.393	2.381	
	Ebn at Noon	799	884	907	886	886	863	832	835	820	808	729	743	
	Edh at Noon	67	87	107	123	119	120	123	111	96	74	65	57	
All-Sky Solar Radiation	RadAvg	1.23	2.20	3.44	4.51	5.20	5.50	5.28	4.51	3.43	2.02	1.08	0.82	
	RadStd	0.14	0.22	0.26	0.38	0.44	0.41	0.52	0.27	0.24	0.17	0.11	0.08	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) +1.87	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (16 neighbors)	DBAvg	(d) N/A	(e) N/A	(f) +1.44	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A

Nomenclature: See separate page